INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

DRILLING MACHINE, ROUND COLUMN, BENCH- AND FLOOR-MOUNTED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers bench- and floor-mounted, round column, drilling machines intended for use in light- to medium-duty general drilling operations, as well as for drilling operations with side-by-side installations of multiple drilling machines on production worktables.

2. CLASSIFICATION. The drilling machines shall be of the following types, styles, and sizes. Size is determined by the drill diameter capacity and the swing (throat dimension). The type, style, and size to be furnished shall be as specified (see 7.3(b)).

Type I - Bench mounted.
Type II - Floor mounted (style A only).

Style A - One spindle.
Style B - Two spindles.
Style C - Three spindles.
Style D - Four spindles.
Style E - Five spindles.
Style F - Six spindles.

Size 1 - 1/2-inch drill diameter, 15-inch swing.
Size 2 - 3/4-inch drill diameter, 17-inch swing.
Size 3 - 1-inch drill diameter, 20-inch swing.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3413

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The drilling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The drilling machine shall include, as a minimum, the following components:

3.2.1 Base. The base of the type I or II, style A drilling machine shall support a worktable, column, and drill head assembly. The base shall have the number and size of T-slots as specified (see 7.3(c)). The dimensions of the T-slots shall conform to the requirements of American Society of Mechanical Engineers (ASME) B5.1M, "T-Slots, Their Bolts, Nuts, and Tongues" (DoD adopted). The base shall have a means for leveling and anchoring the machine to a bench or floor.

3.2.2 Worktable. Unless otherwise specified (see 7.3(d)), the dimensions of the worktable shall be those considered by the manufacturer as standard. The worktable shall rotate 360° about the center of the column and shall be manually adjustable vertically along the column to any position between the base and drill head assembly. A locking device shall be provided to secure the worktable after adjustment. The worktable shall have the number and size of T-slots as specified (see 7.3(e)). The dimensions of the T-slots shall conform to the requirements of ASME B5.1M.

3.2.3 Production worktable. When specified (see 7.3(f)), a production worktable shall be provided for side-by-side installation of type I machines (styles B through F). The worktable shall have a trough around its periphery with provision for draining tool coolant back to the reservoir. Unless otherwise specified (see 7.3(g)), the dimensions of the worktable shall be those considered by the manufacturer as standard for the style of drilling machine specified.

3.2.4 Drill head assembly. The drill head assembly shall include a power transmission system and a quill and spindle. The drill head assembly shall rotate 360° about the center of the column. When specified (see 7.3(h)), the drill head assembly shall be vertically adjustable along the column. A locking device shall be provided to secure the drill head assembly after adjustment.

3.2.4.1 Multiple spindles. Style B through F drilling machines shall have a multiple number of single-spindle machine columns and drill head assemblies mounted side by side on a production worktable. Each column and drill head assembly shall operate as an independent unit, and each unit shall be interchangeable with each of the others. Unless otherwise specified (see 7.3(i)), the spindle center-to-center distance shall be 12, 15, and 17 inches for sizes 1, 2, and 3, respectively.

3.2.4.2 Power transmission. Unless otherwise specified (see 7.3(j)), the spindle speed range and drive motor horsepower shall meet the requirements shown in table I. Unless otherwise specified (see 7.3(k)), power transmission shall be by a belt and step pulley system having at least four spindle speeds for a size 1 machine and five spindle speeds for a size 2 or size 3 machine. When specified (see 7.3(l)), the spindle speed shall be infinitely variable throughout the range.

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TABLE I. Spindle drive requirements.

Machine size	Spindle speed range, minimum, rpm	Motor horsepower, 30-minute duty, minimum
1	500 - 1900	3/4
2	460 - 1800	3/4
3	290 - 1500	1

3.2.4.3 Quill and spindle assembly. Drill feed shall be accomplished by means of a hand-feed spoke lever and a rack and pinion gear arrangement. Unless otherwise specified (see 7.3(m)), the spindle taper of a size 1 or 2 machine shall be a Morse taper No. 2 and of a size 3 machine shall be a Morse taper No. 3. The spindle taper shall meet the requirements of ASME B5.10, "Machine Tapers (Self Holding and Steep Taper Series)" (DoD adopted).

3.2.4.4 Power feed. If required, the spindle shall have power feed within the feed rate ranges as specified (see 7.3(n)).

3.2.4.5 Depth stop. The machine shall have a depth stop equipped with a positive locking device. The depth stop shall be provided with an indicating scale marked in increments of not greater than 1/16 inch.

3.2.4.6 Drill chuck. A drill chuck shall be provided for each spindle with a 1/2-inch diameter drill capacity and taper shank that corresponds to the spindle taper of the size machine specified.

3.2.5 Controls. All controls shall be located for convenient access by the operator. Unless otherwise specified (see 7.3(o)), the following controls, as a minimum, shall be provided:

a. On/off power switch.
b. Emergency stop button.
c. Coolant flow.

3.2.6 Lubrication system. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.7 Coolant system. When specified (see 7.3(p)), a flood tool coolant system shall be provided. As a minimum, the coolant system shall consist of a reservoir, pump, strainer, and all necessary piping. The strainer shall be removable for cleaning. The coolant system shall have a means to control and direct the rate of clean coolant flow within the work area.

3.2.8 Electrical requirements. Unless otherwise specified (see 7.3(q)), the electrical system shall conform to the National Fire Prevention Association (NFPA) 79, "Electrical Standard for Industrial Machinery" (DoD adopted). The machine shall have dual 115/230-volt, single-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.3(r)). Unless otherwise specified (see 7.3(s)), the control system power shall be provided by a separate 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

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3.2.9 Drive motor. The spindle drive motor shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1, "Motors and Generators" (DoD adopted). Unless otherwise specified (see 7.3(t)), the motor shall have open drip-proof enclosure.

3.2.10 Optional accessories. Optional accessories shall be provided as specified (see 7.3(u)).

3.3 Performance. Unless otherwise specified (see 7.3(v)), the drilling machine shall meet the alignment and performance requirements shown in tables II and III, respectively, when tested in accordance with sections 5.3.1 through 5.3.4.

TABLE II. Alignment requirements.

Parameter	Maximum allowable deviation
Base and worktable flatness	0.010 inches
Spindle taper run-out TIR[1] @ 1-1/4 inches from face TIR @ 5 inches from face	 0.002 inches 0.005 inches
Spindle (quill) travel Square with worktable Spindle incline toward column	 0.005 inches 0.005 inches
Worktable travel square to column	0.005 inches

[1] TIR is the total indicated reading.

TABLE III. Performance requirements.

Parameter	Tolerance
Vertical alignment	0.01 inch/foot, TIR[1]
Hole straightness	0.005 inch, TIR hole top to bottom
Hole maximum diameter	Test drill diameter + 0.007 inch
Hole diameter out of roundness	0.005 inch, TIR

[1]TIR is the total indicated reading.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.3(w)).

3.5 Safety and health requirements. The drilling machine shall be designed and manufactured in accordance with American National Standards Institute (ANSI) B11.8, "Drilling, Milling, and Boring Machines - Safety Requirements for the Construction, Care, and Use" (DoD adopted). The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR 1910, "Occupational Safety and Health Standards". If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.3(x)). Regardless of the measurement system used, all measuring and indicating devices on the machine

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shall be graduated in the same system. When specified (see 7.3(y)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.3(z)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power Input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.3(aa)), a lubrication plate or chart shall be attached to the machine. The plate shall include the following information:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The drilling machine provided shall meet the salient characteristics of this commercial item description, conform to the manufacturer's drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The drilling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.3bb)), the machine shall be tested in accordance with sections 5.3.1 through 5.3.4.

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5.3.1 Test setup and conditions. Unless otherwise specified (see 7.3(cc)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.3(dd)). Unless otherwise specified (see 7.3(ee)), the machine shall be tested in accordance with environmental conditions of ambient temperature between 40 °F and 104 °F and humidity between 20 and 95 percent, noncondensing.

5.3.2 Operational test. The drilling machine shall be operated in accordance with the manufacturer's standard operating procedures. The drilling machine shall be operated at no load for not less than 1/2 hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The performance tests shall be conducted on the drilling machine in accordance with section 5.3.3.2. There shall be no evidence of overloading, overheating, excessive vibration, or deformation of part of the machine during or after the test.

5.3.3.1 Machine alignment. A machine alignment inspection shall be performed before and after the performance test using the criteria shown in table II. Changes in machine alignment shall be cause for rejection.

5.3.3.2 Drill test. A test workpiece 1.0 inch in thickness and made of soft cast iron with a Brinell hardness of 120 to 150 shall be used. The test shall consist of drilling 5 holes through the test workpiece in 12 minutes. The drill bit nominal diameter shall be the maximum rated drilling diameter for the machine. Drill bit material, drill spindle speed, and feed rate shall be at the option of the manufacturer. The holes shall be measured and shall not exceed the values shown in table III for vertical alignment, hole straightness, maximum hole diameter, and out of roundness. The diameter of all holes shall be measured from the top to the bottom in three places. Out of roundness (the difference between the largest diameter of the hole and the smallest diameter of the hole) shall be determined by moving the measuring device 360° around the drilled hole.

5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.3(ff)).

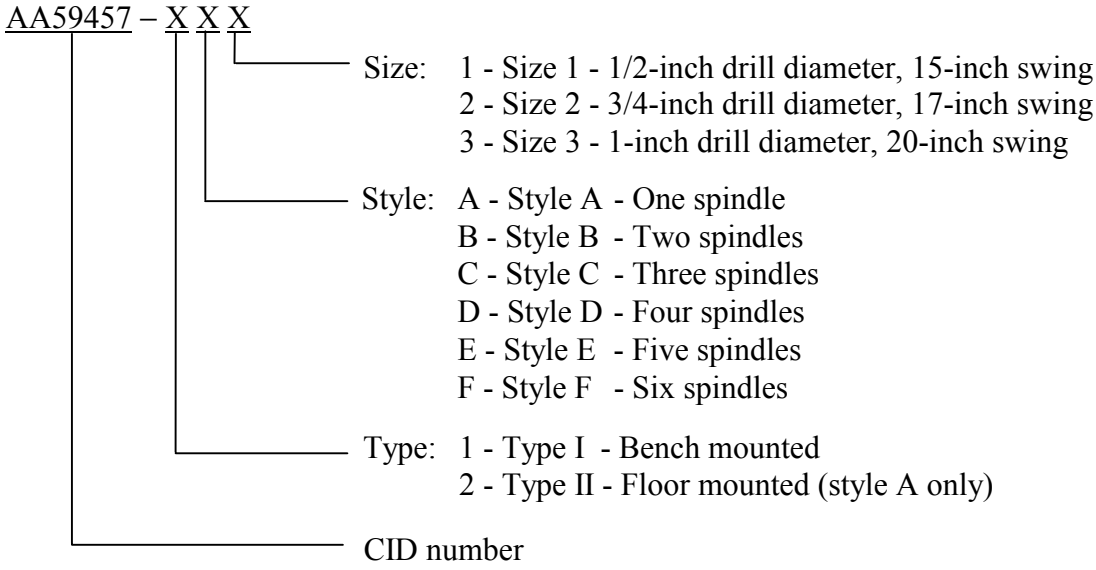
5.4 Acceptance. Unless otherwise specified (see 7.3(gg)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables II and III. Failure of the machine to meet these requirements shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.3(hh)), preservation, packing, and marking shall be as specified in the contract or order.

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7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



7.2 Sources of documents.

7.2.1 FAR. A copy of the FAR may be obtained from the U.S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328.

7.2.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.2.3 ASME standards. Copies of ASME standards may be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

7.2.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.2.5 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

7.2.6 OSHA standards. Copies of OSHA standards may be obtained from U.S. Department of Labor, 200 Constitution Avenue NW, Room N-3649, Washington, DC 20210. Electronic copies of OSHA standards may be obtained from http:/www.osha.gov.

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7.3 Ordering data. Acquisition documents must specify the following:

a. CID document number, revision, and CID PIN.
b. Type, style, and size of drilling machine required (see 2).
c. Base number and size of T-slots, as specified (see 3.2.1).
d. Worktable dimensions, if different (see 3.2.2).
e. Worktable number and size of T-slots, as specified (see 3.2.2).
f. Production worktable, if required (see 3.2.3).
g. Production worktable dimensions, if different (see 3.2.3).
h. Vertically adjustable drill head assembly, if required (see 3.2.4).
i. Multiple-spindle center-to-center distance, if different (see 3.2.4.1).
j. Spindle speed range and drive motor horsepower, if different (see 3.2.4.2).
k. Power transmission system, if different (see 3.2.4.2).
l. Infinitely variable spindle speed, if required (see 3.2.4.2).
m. Spindle taper, if different (see 3.2.4.3).
n. Spindle power feed, as specified (see 3.2.4.4).
o. Controls, if different (see 3.2.5).
p. Coolant system, if required (see 3.2.7).
q. Electrical system industry standard, if different (see 3.2.8).
r. Electrical system voltage, as specified (see 3.2.8).
s. Control system voltage, if different (see 3.2.8).
t. Electric motor enclosure, if different (see 3.2.9).
u. Optional accessories, as specified (see 3.2.10).
v. Performance, if different (see 3.3).
w. Machine dimensions and weight, as specified (see 3.4).
x. System of units required (see 3.6).
y. Dual system of units, if required (see 3.6).
z. Nameplate, if different (see 3.7).
aa. Lubrication plate or chart, if required (see 3.8).
bb. Testing, if different (see 5.3).
cc. Test responsibility, if different (see 5.3.1).
dd. Certification of calibration, as specified (see 5.3.1).
ee. Test environmental conditions, if different (see 5.3.1).
ff. Optional tests, as specified (see 5.3.4).
gg. Acceptance, if different (see 5.4).
hh. Packaging requirements, if different (see 6).

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - SH
Air Force - 99

Reviewers:
Army - SM
Navy - YD
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FEE

Preparing Activity:
DLA - GS6

(Project 3413-0169)

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